Assignment Instructions

PS: READ CAREFULLY THE INSTRUCTIONS NFW AUTOMATIC DEDUCTION OF POINTS!

IT15 LAB EXAM

ASP.NET Core CRUD Application (Employee & Payroll System)

Scenario

You have been hired as a junior developer for a small company called **TechCore Solutions**. The HR department currently manages employee records and payroll manually using spreadsheets, which often leads to errors and delays.
Your task is to develop a **web-based Employee Management and Payroll System** using **ASP.NET Core (C# only)** to help automate their workflow.
The system should allow HR staff to:

- Manage employee records
- Process payroll transactions
- View salary computations

Objective

Build a **CRUD-based ASP.NET Core web application** that manages:

1. Employee Information
2. Payroll Transactions

Technical Requirements

- Framework: **ASP.NET Core MVC**
- Language: **C# only**
- Database: **SQL Server / Xampp phpmyadmin**
- ORM: **Entity Framework Core**
- UI: Basic Razor Views (no frontend frameworks required)

System Features

Employee Module (CRUD)

Create a module that allows:

- Add Employee
- View All Employees
- Edit Employee
- Delete Employee

Employee Fields:

- Employee ID (Auto-generated)
- First Name
- Last Name
- Position
- Department
- Daily Rate

Payroll Module (CRUD)

Each employee can have multiple payroll transactions.

- Add Payroll Record
- View Payroll History per Employee
- Edit Payroll
- Delete Payroll

Payroll Fields:

- Payroll ID
- Employee ID (Foreign Key)
- Date
- Days Worked
- Gross Pay (Days Worked × Daily Rate)
- Deduction
- Net Pay (Gross Pay - Deduction)

Business Logic Requirements

- Automatically compute:
- Gross Pay
- Net Pay
- Validate inputs (e.g., no negative values for days worked or deductions)
- Ensure relational integrity (Employee ↔ Payroll)

Interface Requirements

- Simple and clean UI

- Navigation menu:
- Employees
- Payroll
- Use tables to display records
- Use forms for input

Deliverables

1. GitHub Repository

- Push your project to GitHub
- Repository name format:

```
IT15-LabExam-Lastname
```

- Must include:
- Complete source code
- Database migration files

2. Screenshots (Compile into ONE PDF or DOCX)

Include screenshots of:

1. Employee List Page
2. Add Employee Form
3. Edit Employee
4. Payroll List per Employee
5. Add Payroll Transaction
6. Computed Payroll (Gross & Net Pay visible)
7. Database table structure (optional but bonus)

Restrictions

- No use of other programming languages (C# only)
- No copy-paste from classmates
- No use of full templates (you must build your own structure)
- No frontend frameworks (Bootstrap allowed)

Grading Criteria (100 pts)

Criteria	Points
Employee CRUD	25

Payroll CRUD	25
Business Logic (Computation)	20
UI & Usability	10
Code Quality	10
GitHub Submission	5
Screenshots	5

Submission

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